

Array Programming

Interview Question Bank

60 Must-Know Questions | Easy to Advanced | Google, Amazon, Microsoft & More



Overview

60 Total Questions	20 Easy	20 Intermediate	20 Hard
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Target Companies

Google	Amazon	Microsoft	Adobe	Uber	Flipkart	TCS	Infosys	HoneyBell
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How to Use This Sheet

Step 1 — Start with Easy Master all 20 basics questions first. Focus on understanding time and space complexity for each.	Step 2 — Move to Intermediate Learn Two Pointers, Sliding Window, Kadane's, Sorting and Binary Search patterns thoroughly.	Step 3 — Conquer Hard Advanced problems combine multiple patterns. Understand the approach before coding.
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SECTION 1 — EASY (Questions 1–20)

Focus: Pure array manipulation, no special algorithm required. Build intuition for traversal patterns, index logic, and basic data structures.

No.	Question	Topic / Pattern	Companies	Key Approach	Time	Space
1	Find the maximum	Basics	TCS, Infosys,	Linear scan	O(n)	O(1)

No.	Question	Topic / Pattern	Companies	Key Approach	Time	Space
	and minimum element in an array		Wipro	with two variables		
2	Reverse an array in place	Basics	TCS, Infosys, Amazon	Two pointers from both ends	O(n)	O(1)
3	Find the sum and average of all elements	Basics	TCS, Wipro	Single pass summation	O(n)	O(1)
4	Count the frequency of each element	Basics	Infosys, Adobe	HashMap frequency count	O(n)	O(n)
5	Check if an array is sorted (asc or desc)	Basics	TCS, HoneyBell	Compare adjacent pairs	O(n)	O(1)
6	Remove duplicates from a sorted array	Basics	Amazon, Flipkart, Google	Slow pointer technique	O(n)	O(1)
7	Find the second largest element	Basics	TCS, Infosys, Microsoft	Single pass, two variables	O(n)	O(1)
8	Rotate an array by K positions (left/right)	Basics	Amazon, Adobe, Flipkart	Reverse trick (3 reversals)	O(n)	O(1)
9	Move all zeros to the end (preserve order)	Basics	Microsoft, Adobe, Uber	Two pointers	O(n)	O(1)
10	Find all pairs that sum to K	Basics	Amazon, Google, Uber	HashSet lookup	O(n)	O(n)
11	Find the intersection of two arrays	Basics	Microsoft, Amazon, Adobe	HashSet or two pointers	O(n)	O(n)
12	Find the union of two arrays	Basics	TCS, Infosys	HashSet merge	O(n)	O(n)
13	Check if two arrays are equal	Basics	TCS, Wipro	Sort and compare	O(n log n)	O(1)
14	Find the missing number in array 1 to N	Math / XOR	Google, Amazon, Microsoft, Flipkart	Sum formula: $N*(N+1)/2$ - sum	O(n)	O(1)

No.	Question	Topic / Pattern	Companies	Key Approach	Time	Space
15	Find the duplicate in array of 1 to N	Math / XOR	Amazon, Google, Adobe	XOR or sum formula	O(n)	O(1)
16	Find majority element (>N/2 times) — brute force	Frequency	TCS, Infosys, Flipkart	HashMap count	O(n)	O(n)
17	Range sum query (sum between L and R)	Prefix Sum	Amazon, Google, Microsoft	Prefix array: P[R] - P[L-1]	O(1)/query	O(n)
18	Find the equilibrium index	Prefix Sum	Adobe, Amazon, Infosys	Prefix sum comparison	O(n)	O(1)
19	Count even and odd numbers	Basics	TCS, Wipro	Modulo check in one pass	O(n)	O(1)
20	Largest element in each subarray of size K (brute)	Sliding Window	Amazon, Flipkart	Nested loops over windows	O(nk)	O(1)

Pattern Cheatsheet — Easy

Pattern	When to Use	Classic Example
Linear Scan	Find max/min, count, frequency	Find max element, count frequency
Two Pointers	Reversal, pair sum, sorted merge	Reverse array, find pair with sum K
HashSet / HashMap	Duplicates, intersections, lookups	Intersection, missing number
Prefix Sum	Range queries, equilibrium index	Sum between L and R in O(1)

SECTION 2 — INTERMEDIATE (Questions 21–40)

Focus: Algorithmic patterns — Two Pointers, Sliding Window, Sorting, Binary Search, and Kadane's Algorithm. Each question maps to a specific reusable pattern.

No.	Question	Topic / Pattern	Companies	Key Approach	Time	Space
21	3Sum — find all triplets summing to zero	Two Pointers	Google, Amazon, Microsoft, Adobe, Uber	Sort + fix one + two pointers	$O(n^2)$	$O(1)$
22	Container with most water	Two Pointers	Google, Amazon, Microsoft, Uber, Flipkart	Two pointers from both ends	$O(n)$	$O(1)$
23	Merge two sorted arrays without extra space	Two Pointers	Amazon, Microsoft, Flipkart, Adobe	End-of-first, start-of-second pointer	$O(n+m)$	$O(1)$
24	Longest subarray with sum equal to K (with negatives)	Two Pointers + HashMap	Amazon, Flipkart, Google	Prefix sum + HashMap	$O(n)$	$O(n)$
25	Remove all occurrences of value in-place	Two Pointers	Microsoft, Amazon	Slow/fast pointer write-over	$O(n)$	$O(1)$
26	Sort array of 0s, 1s, 2s (Dutch National Flag)	Sorting	Google, Amazon, Microsoft, Adobe, Flipkart	3 pointers: low, mid, high	$O(n)$	$O(1)$
27	Kth largest/smallest element in unsorted array	Sorting / Heap	Google, Amazon, Microsoft, Uber, Adobe	QuickSelect or min-heap of size k	$O(n)$ avg	$O(1)$
28	Implement merge sort from scratch	Sorting	Google, Microsoft, Amazon	Divide and merge recursively	$O(n \log n)$	$O(n)$
29	Count inversions in an array	Sorting	Google, Amazon, Adobe,	Modified merge sort	$O(n \log n)$	$O(n)$

No.	Question	Topic / Pattern	Companies	Key Approach	Time	Space
			Flipkart			
30	First and last position of target in sorted array	Binary Search	Google, Amazon, Microsoft, Uber	Two binary searches	$O(\log n)$	$O(1)$
31	Search in a rotated sorted array	Binary Search	Google, Amazon, Microsoft, Adobe, Uber	Detect sorted half, narrow down	$O(\log n)$	$O(1)$
32	Find the peak element	Binary Search	Google, Amazon, Microsoft	Compare mid and mid+1	$O(\log n)$	$O(1)$
33	Square root of a number using binary search	Binary Search	Amazon, Microsoft, Adobe	Binary search on answer space	$O(\log n)$	$O(1)$
34	Maximum sum contiguous subarray (Kadane's)	Kadane's	Google, Amazon, Microsoft, Adobe, Flipkart, Uber	maxEndingHere = max(arr[i], prev+arr[i])	$O(n)$	$O(1)$
35	Maximum product contiguous subarray	Kadane's	Google, Amazon, Microsoft, Adobe	Track both max and min product	$O(n)$	$O(1)$
36	Maximum sum subarray of size K	Sliding Window	Amazon, Microsoft, Adobe, Flipkart	Add right, remove left element	$O(n)$	$O(1)$
37	Longest subarray with at most K distinct elements	Sliding Window	Google, Amazon, Uber	HashMap + two pointers	$O(n)$	$O(k)$
38	Smallest subarray with sum $\geq S$	Sliding Window	Amazon, Microsoft, Flipkart	Expand right, shrink left	$O(n)$	$O(1)$
39	Maximum sum of exactly-K size subarray (prefix)	Prefix Sum	Amazon, Infosys, TCS	prefix[i] - prefix[i-k]	$O(n)$	$O(n)$

No.	Question	Topic / Pattern	Companies	Key Approach	Time	Space
40	Find the celebrity in a group of N people	Sorting / Elimination	Amazon, Microsoft, Adobe, Flipkart	Eliminate with one pass, verify	O(n)	O(1)

Pattern Cheatsheet — Intermediate

Pattern	Time	Key Idea	Classic Problems
Two Pointers	O(n)	Left + right converge or move together	3Sum, Container Water, Merge Sorted
Sliding Window	O(n)	Expand right, contract left on violation	Max sum K-window, Smallest subarray
Binary Search	O(log n)	Eliminate half the search space each step	First/last pos, Rotated array, Peak
Kadane's Algorithm	O(n)	Keep running sum, reset on negative total	Max subarray sum, Max product
Sorting	O(n log n)	Sort enables two-pointer / binary search	Dutch Flag, Kth largest, Inversions

SECTION 3 — HARD (Questions 41–60)

Focus: Advanced interview questions combining multiple patterns. These appear frequently at Google, Amazon, Microsoft, and Uber final rounds.

No.	Question	Topic / Pattern	Companies	Key Approach	Time	Space
41	Trapping rainwater	Advanced	Google, Amazon,	Prefix max +	O(n)	O(1)

No.	Question	Topic / Pattern	Companies	Key Approach	Time	Space
	— total water trapped		Microsoft, Adobe, Uber, Flipkart	suffix max arrays		
42	Sliding window maximum (max in every K-window)	Monotonic Deque	Google, Amazon, Microsoft, Uber	Monotonic decreasing deque	$O(n)$	$O(k)$
43	Longest consecutive sequence (unsorted array)	HashSet	Google, Amazon, Uber, Flipkart	HashSet; count only from sequence start	$O(n)$	$O(n)$
44	Elements appearing $> N/3$ times (extended vote)	Boyer-Moore	Google, Amazon, Adobe, Microsoft	Two candidates with Boyer-Moore	$O(n)$	$O(1)$
45	Count subarrays with XOR equal to K	XOR + HashMap	Amazon, Flipkart, Google, Uber	Prefix XOR + HashMap lookup	$O(n)$	$O(n)$
46	Merge overlapping intervals	Intervals	Google, Amazon, Microsoft, Adobe, Uber, Flipkart	Sort by start, merge greedily	$O(n \log n)$	$O(n)$
47	Minimum platforms needed at a station	Intervals	Amazon, Flipkart, Microsoft, Adobe	Sort arrivals & departures, sweep	$O(n \log n)$	$O(1)$
48	Jump game — can you reach the last index?	Greedy	Google, Amazon, Uber, Flipkart	Track maxReach greedily	$O(n)$	$O(1)$
49	Jump game II — minimum jumps to last index	Greedy	Google, Amazon, Microsoft, Uber	Track window end & farthest reach	$O(n)$	$O(1)$
50	Stock buy/sell — at most one transaction	Greedy	Google, Amazon, Microsoft, Adobe, Uber, Flipkart	Track min price, compute profit	$O(n)$	$O(1)$
51	Stock buy/sell — unlimited transactions	Greedy	Amazon, Adobe, Flipkart	Sum all uphill differences	$O(n)$	$O(1)$

No.	Question	Topic / Pattern	Companies	Key Approach	Time	Space
52	Median of two sorted arrays	Binary Search	Google, Amazon, Microsoft, Uber	Binary search on smaller array partitions	$O(\log \min(m,n))$	$O(1)$
53	Count subarrays with product less than K	Sliding Window	Google, Amazon, Uber, Adobe	Two pointers multiply/divide	$O(n)$	$O(1)$
54	Longest subarray with equal 0s and 1s	Prefix Sum + HashMap	Amazon, Google, Flipkart, Adobe	Replace 0 $\rightarrow$ -1, find longest sum=0	$O(n)$	$O(n)$
55	Count subarrays with sum equal to K	Prefix Sum + HashMap	Google, Amazon, Microsoft, Uber, Flipkart	freq[prefix - K] lookup	$O(n)$	$O(n)$
56	Spiral order traversal of a 2D matrix	Matrix	Google, Amazon, Microsoft, Adobe	Four boundary pointers, peel layers	$O(m*n)$	$O(1)$
57	Set matrix zeros (row & column to 0)	Matrix	Google, Amazon, Microsoft, Adobe	Use first row/col as markers	$O(m*n)$	$O(1)$
58	Rotate a matrix 90 degrees in place	Matrix	Google, Amazon, Microsoft, Adobe, Uber	Transpose then reverse each row	$O(n^2)$	$O(1)$
59	Minimum size subarray sum $\geq S$ — $O(n)$	Sliding Window	Amazon, Microsoft, Google, Uber	Expand right, contract left window	$O(n)$	$O(1)$
60	Largest rectangle in histogram	Stack / Advanced	Google, Amazon, Microsoft, Adobe, Uber	Monotonic stack, track width/height	$O(n)$	$O(n)$

Pattern Cheatsheet — Hard

Pattern	Time	Key Idea	Classic Problems
Prefix Sum +	$O(n)$	Count subarrays via running	Sum=K, XOR=K, Equal 0s

Pattern	Time	Key Idea	Classic Problems
HashMap		total lookup	and 1s
Monotonic Deque	$O(n)$	Maintain useful elements in decreasing order	Sliding window maximum
Greedy	$O(n)$	Locally optimal $\Rightarrow$ globally optimal	Jump game I & II, Stock profit
Boyer-Moore Voting	$O(n)$	Cancel votes to find majority candidates	$N/2$ majority, $N/3$ majority
Interval Merge	$O(n \log n)$	Sort by start, merge overlapping ranges	Merge intervals, Platforms
Matrix Manipulation	$O(n^2)$	Boundary/index tricks, in-place transforms	Spiral, Rotate 90, Set Zeros

Recommended Study Roadmap

Week	Phase	Questions	Patterns	Goal
1-2	Easy	Q1-20	Scan, HashMap, Prefix Sum	Build array intuition
3-4	Intermediate I	Q21-30	Two Pointers, Sorting	Master $O(n)$ patterns
5-6	Intermediate II	Q31-40	Binary Search, Sliding Window, Kadane's	Speed and accuracy
7-8	Hard	Q41-60	All advanced patterns	Interview readiness